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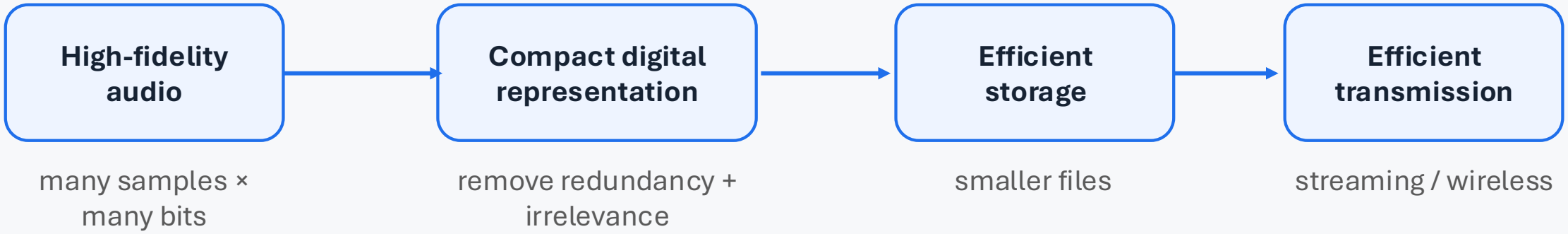
Introduction to audio coding

based on Spanias, Painter & Atti, Audio Signal Processing and Coding, Chapter 1

Core question: how can we reduce bits without changing what listeners perceive?

Why do we need audio coding?

Digital audio is high quality, but raw PCM consumes bandwidth and storage.



Audio coding is the engineering compromise between rate, quality, delay, and robustness.

Learning goals for this introduction

1. Compute raw PCM bit rates from fs, bit depth, and channels.
2. Explain why sampling and quantization are not compression by themselves.
3. Distinguish lossless, perceptual, parametric, and neural codecs.
4. Read a generic perceptual encoder block diagram.
5. Evaluate codecs using quality, bit rate, complexity, delay, and robustness.

The course at a glance

Six topics, from representation and numerical precision to modern codec architectures.

01 Introduction & PCM

Sampling, bitrate, coding goals

02 Quantization & entropy

Bit allocation, distortion, Huffman

03 Time–frequency analysis

FIR/IIR, filter banks, DFT/DCT/MDCT

04 Psychoacoustics

Critical bands, masking, NMR/SMR

05 Audio coding standards

MP3, AAC, Opus, Atmos, MPEG-H, FLAC

06 Neural compression

CNNs, autoencoders, RVQ, EnCodec

Recurring question: which information needs bits—and how can we encode it efficiently?

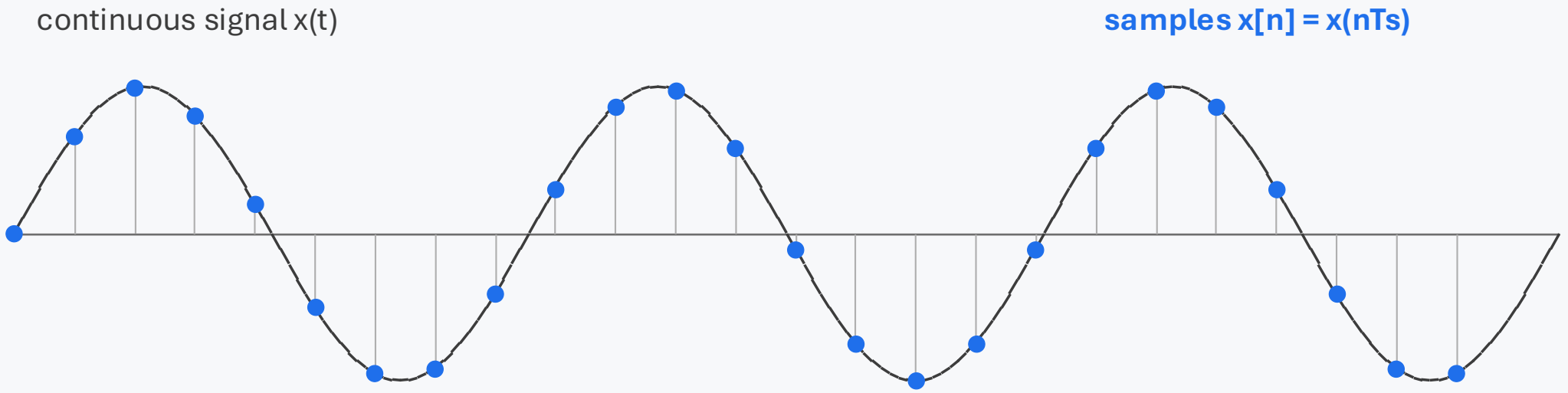
From acoustic pressure to a stream of bits



Sampling discretizes time. Quantization discretizes amplitude. Coding reduces the number of bits needed to represent the signal.

Sampling: time becomes an index

Uniform sampling takes values of a continuous signal at regularly spaced times.



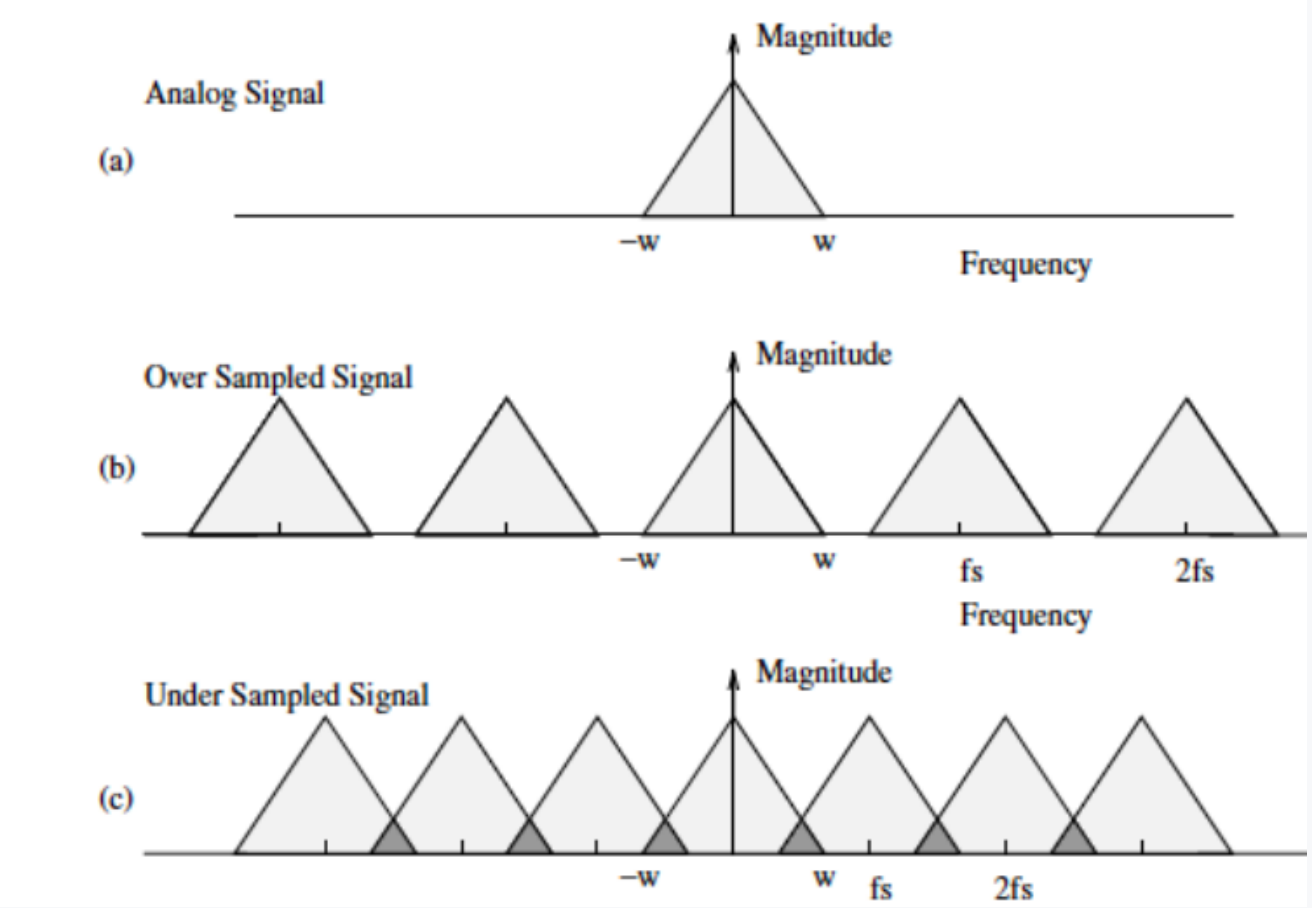
fs = 1 / Ts

t = nTs = n / fs

CD audio: 44,100 samples/s

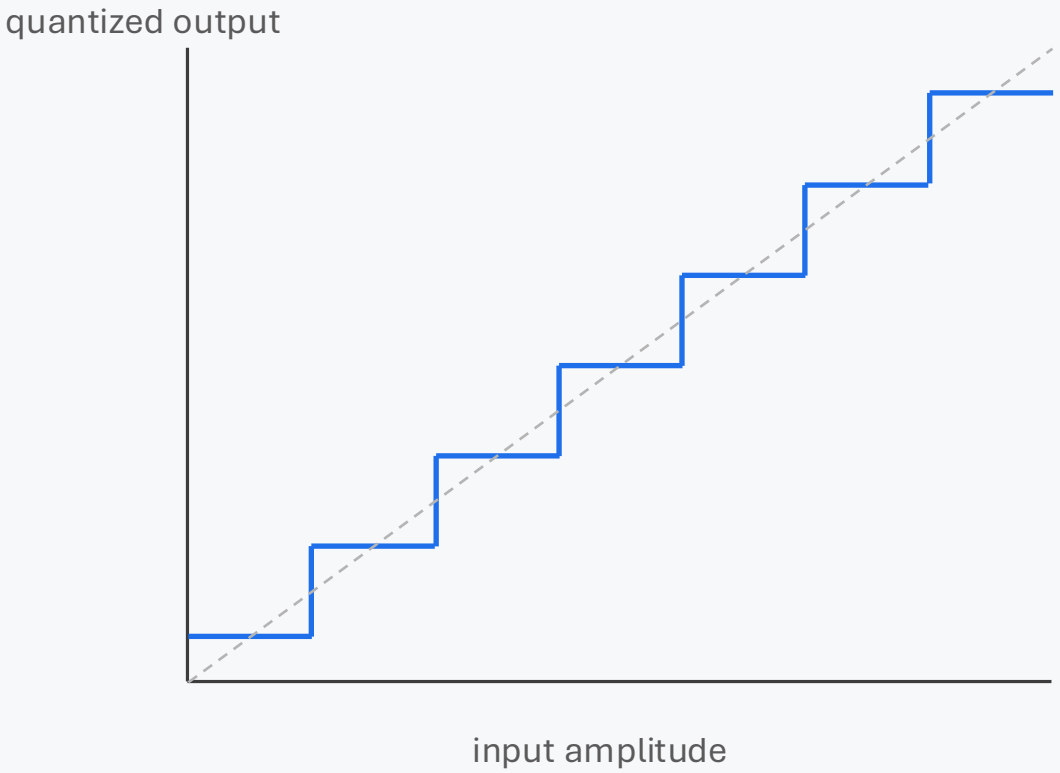
Sampling theorem: avoid spectral overlap

Exact reconstruction is possible when fs > 2 fmax. If spectral replicas overlap, aliasing occurs.



Quantization: amplitude becomes a finite set

A B-bit quantizer maps each sample amplitude to one of 2^B discrete levels.



More bits → more levels → lower quantization noise

But raw PCM rate grows linearly with bit depth B

Perceptual coders place noise where it is least audible

Raw PCM bit rate: the basic calculation

bit rate = channels × bit depth × sampling rate

Audio CD	→	2 channels	→	16 bits	→	44.1 kHz

2 × 16 × 44,100 = 1,411,200 bit/s = 1,411.2 kbit/s

What can an audio coder remove?

Statistical redundancy
Predictable structure in the signal

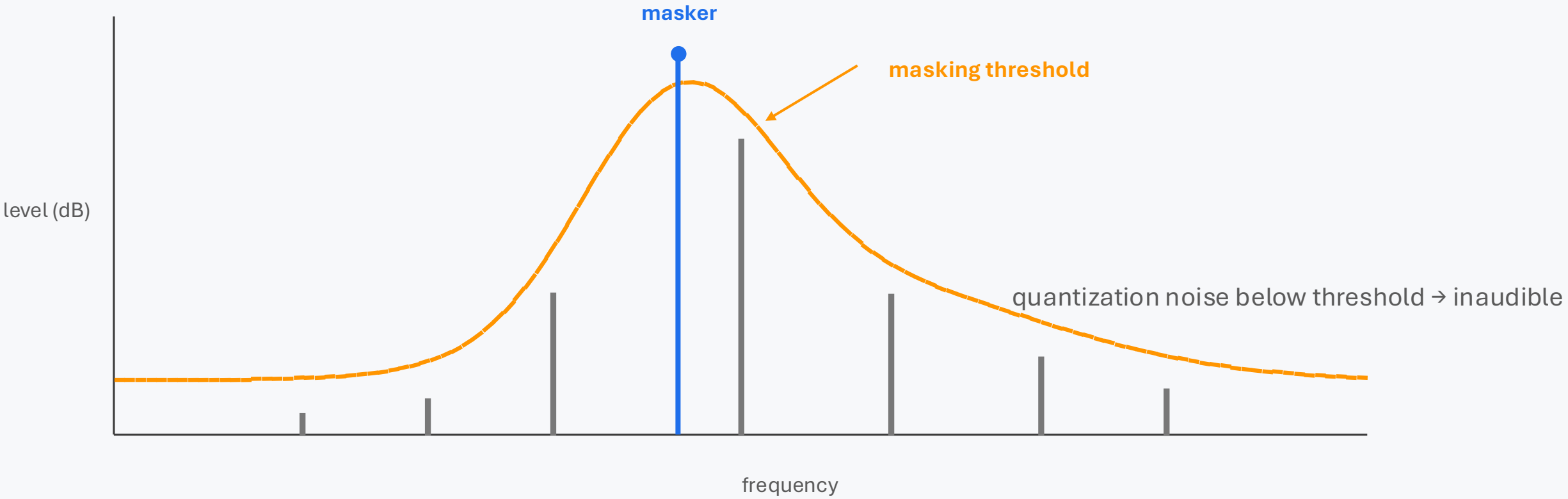
Perceptual irrelevance
Signal components or noise that listeners do not notice

Lossless coding removes redundancy only.
Lossy perceptual coding also exploits irrelevance.

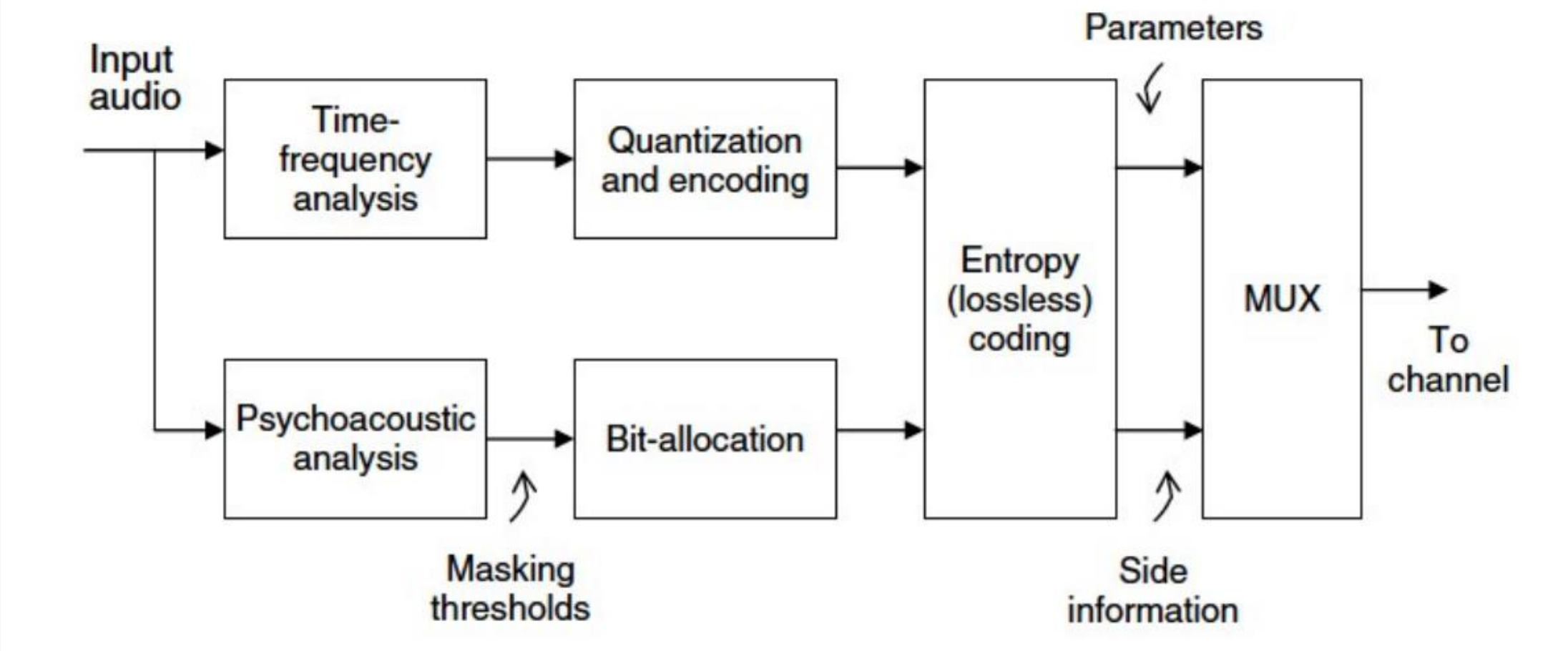
This is why lossy coders can reach much higher compression ratios, at the cost of perfect reconstruction.

The perceptual coding principle

Quantization noise is allowed, but it should stay below the masking threshold.



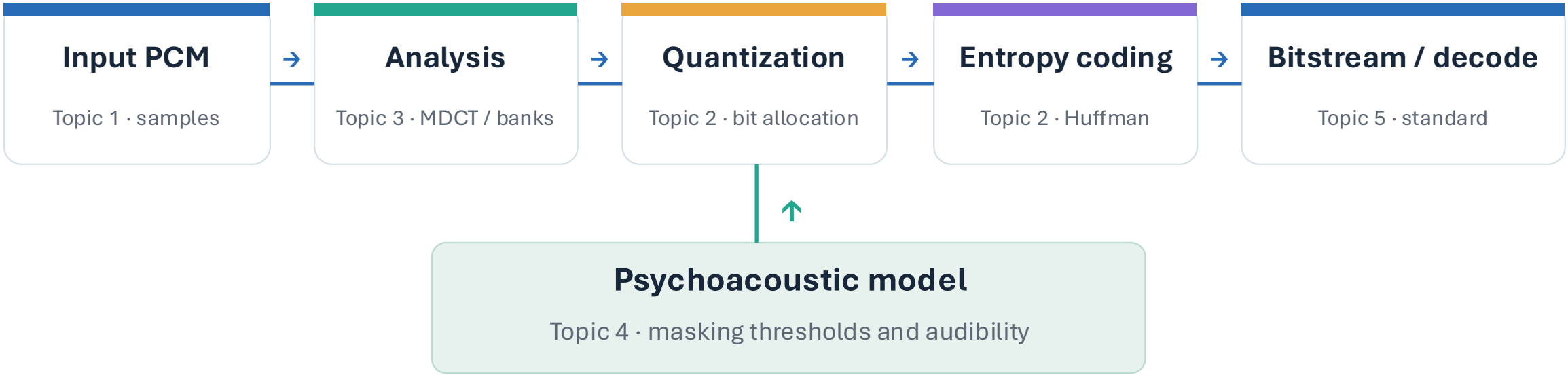
Generic perceptual audio encoder



The psychoacoustic model controls quantization through masking thresholds; entropy coding removes remaining statistical redundancy.

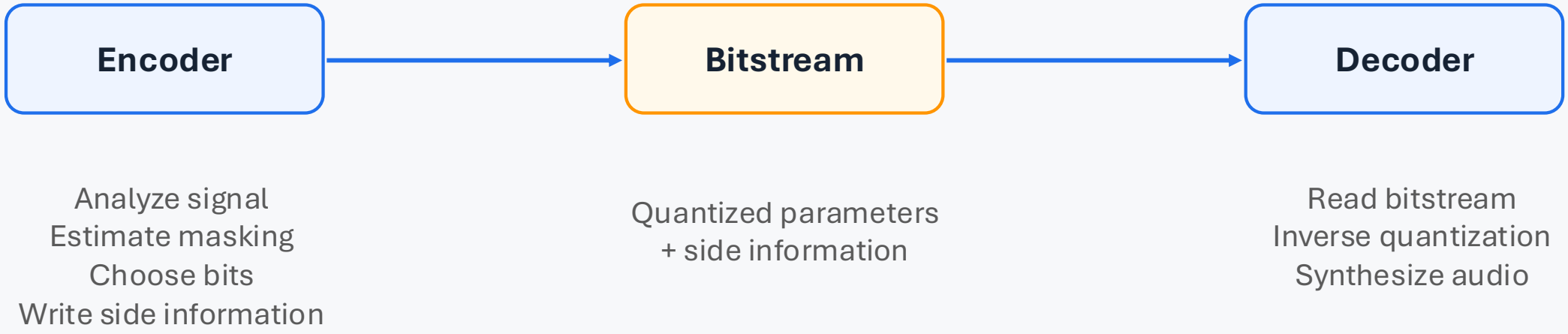
The codec architecture connects the topics

An encoder combines time–frequency analysis, bit allocation and coding, informed by human perception.



Topic 6: neural systems learn analysis and synthesis, using discrete codes such as RVQ.

Encoding and decoding are asymmetric



Most complexity is usually in the encoder, where analysis, psychoacoustics and bit allocation are performed.

Audio coder attributes

Quality

Can listeners hear artifacts?

Bit rate

How many kb/s?

Complexity

Can it run in real time?

Delay

Is it usable for interaction?

Robustness

What happens with losses/errors?

Codec choice is application-dependent: music streaming, broadcasting, storage, games, hearing devices, and real-time communication optimize different tradeoffs.

Main families of audio coders

Lossless

Exact reconstruction
Example: FLAC

Predictive

Exploit short/long-term correlation
Often speech-oriented

Transform/subband

Time-frequency coefficients
Used in many MPEG coders

Sinusoidal/parametric

Represent sound by model parameters

Neural

Learned analysis/synthesis and quantization

What is actually represented and coded?

Modern systems differ in what they store, transmit and reconstruct—not just in their bitrates.

Lossless waveform

PCM prediction / residuals

FLAC

Exact sample recovery

Perceptual signal

Quantized spectral or predictive data

MP3 · AAC · Opus

Masking-guided approximation

Immersive scene

Audio plus objects and spatial metadata

Atmos · MPEG-H

Rendered for the playback device

Learned tokens

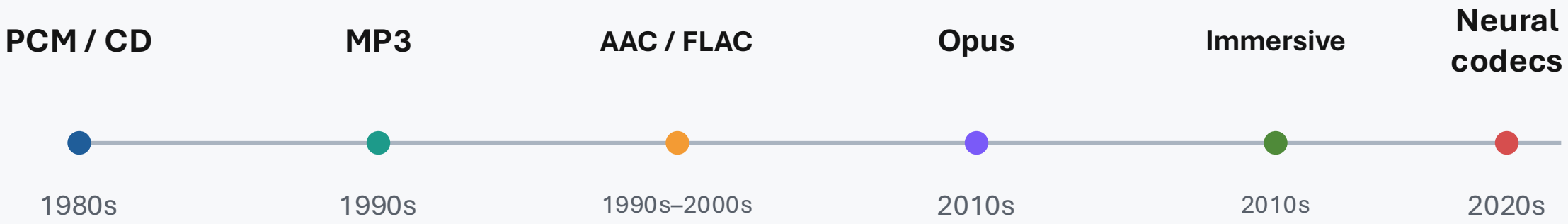
Indices of learned quantization codebooks

SoundStream · EnCodec

Neural decoder reconstructs audio

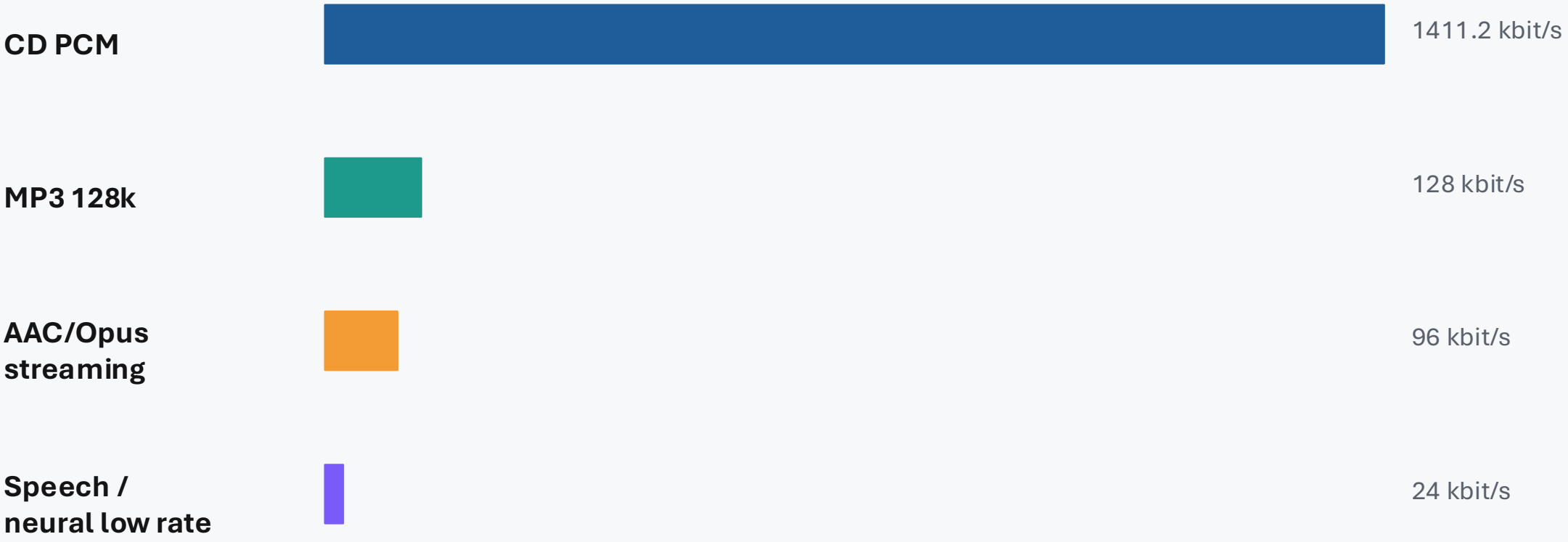
Important: immersive scene representation is distinct from the codec carrying its audio.

Standards and formats: a historical arc



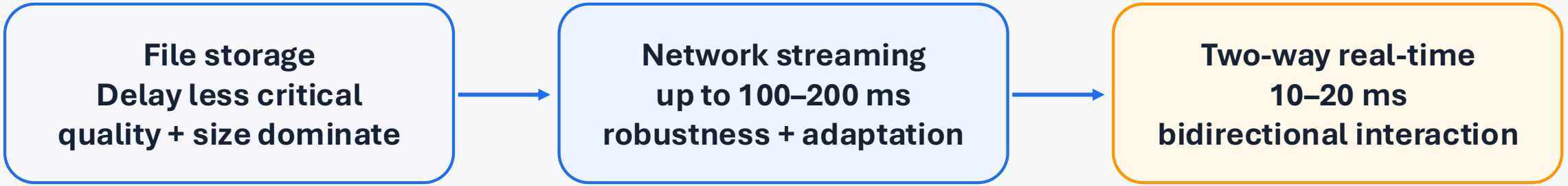
Coding progressed from PCM and transform-based compression to lossless, low-delay, immersive and learned representations.

Bit-rate ladder: compression as a scale



Compression ratio vs CD PCM: MP3 128 kbit/s is about 11:1

Delay matters: choose the codec for the use case



Lower delay usually means shorter frames and less look-ahead, which can reduce coding efficiency.

Which technology suits which application?

There is no universal best codec: choose according to constraints and required representation.

Application	Example technology	Main reason
Music archive / master	**FLAC**	Exact PCM; lossless compression
Music and video streaming	**AAC · xHE-AAC**	Efficient lossy coding over varied bitrates
Interactive calls / conferencing	**Opus**	Low latency; adaptable speech and music
Immersive playback	**Dolby Atmos · MPEG-H**	Channel/object scene plus spatial rendering
Research on learned coding	**SoundStream · EnCodec**	Learned analysis, RVQ, neural synthesis

Compare all options using rate, perceived quality, delay, complexity, robustness and compatibility.

Suggested class demos

Using the examples in https://github.com/xserra/audio-coding-materials/

1	**sound-display.ipynb**	Display a waveform and listen to the relation between samples and sound.
2	**downsample-chirp.ipynb**	Hear and visualize aliasing when a chirp is downsampled.
3	**resample.ipynb**	Compare resampling choices and their effect on spectral content.
4	**mp3-coding.ipynb**	Encode audio as MP3 and compare bit rate, file size and artifacts.
5	**encodec.ipynb**	Compare original and decoded audio from a neural codec (Topic 6).

Takeaways

PCM defines the rate baseline; audio coding reduces bits under quality and delay constraints.

Quantization controls distortion; entropy coding removes statistical redundancy.

Filter banks / MDCT and psychoacoustic masking guide classical perceptual coding.

MP3, AAC, Opus, FLAC and immersive systems make different engineering trade-offs.

Neural codecs (SoundStream / EnCodec) replace fixed analysis with learned representations and RVQ.